

Freemium V1.0 — Manual Test Plan

A scenario-based test plan for the V1.0 freemium model, designed to use the in-app **Debug menu** (Settings → Debug) so a single tester can land every state without manually creating dozens of customers or burning real Vertex AI calls.

Each section is one feature. Each row is: **Setup** → **Action** → **Expected result** → **Pass?**

Before you start

- Run on an Android emulator OR an iPhone simulator. iOS sim is preferred since it exercises the GitLive Firebase Kotlin SDK path that's bitten us most.
- Sign in as one of the test accounts (Fola / Gabby) before running any scenario.
- The freemium server function (`smartDraftMessage`) must be deployed with the post-V1.0-review fix — see [Prerequisites](#).

Quick reference — every debug-menu action

Section	Action	What it does	Use it to test
Seed data	Brand-new tailor	Wipes all customer/order data	Reset to empty state
	Active workshop	8 customers, 4 measurements, 4 orders (deterministic IDs)	Realistic populated dashboard
	All-reconnect	6 customers all >100 days inactive	Reconnect NBA card
	Clear active state	Clears the "Active scenario" pill	Cosmetic only
Bulk seed	Seed N demo customers...	Adds N "Demo Customer #" rows (additive — does not wipe), with optional measurements + orders	Welcome 30-cap, post-welcome 15-cap, swap, slot lock
Session	Reset onboarding flags	Re-shows onboarding + workshop setup on next launch	Onboarding QA
	Sign out	Signs out + returns to login	Account switch, cache reset
Switch account	Switch to Fola / Gabby	Hot-swaps to the other test account	Cross-account regression
Freemium · tier	Set tier: Free / Pro / Atelier	Writes subscriptionTier on user doc	Tier-derived caps + Smart quotas
Freemium · welcome window	Expire welcome window	Backdates welcomeBonusAppliedAt by 35 days → Free 15-cap takes effect	Post-welcome state

	Reset welcome window (now)	Sets welcomeBonusAppliedAt to now + refills bonus coins to 30	Fresh welcome state
Freemium • Smart coins	Drain bonus coins (→ 0)	Sets bonusCoins on user doc to 0 (user-doc field only)	Bonus drained, on-doc reads
	Refill bonus coins (→ 30)	Sets bonusCoins on user doc back to 30	Re-arm welcome bonus path
	Reset Smart usage doc	Deletes users/{uid}/usage/smart_drafts entirely	Brand-new Smart user simulation
	Set Smart usage...	Writes count + bonusBalance directly to the usage doc	Land at any Smart state without burning real AI calls
Freemium • slots	Reconcile customer slots	Calls the reconcileSlots cloud function	Trigger slot relock after cap shrinks
Danger zone	Wipe my data	Wipes ALL customers + orders + measurements + styles for this user	Hard reset between scenarios

Prerequisites

Before running any Smart-draft scenario:

```
cd functions && firebase deploy --only functions:smartDraftMessage
```

This is required because the deployed `smartDraftMessage` was missing the `limit` fallback fix from commit `b6bec0b`. Without redeploying, **every** Smart draft attempt fails with `Cannot use "undefined" as a Firestore value (found in field "limit")`. Confirm the deploy succeeded by looking at the function version in the Firebase Console.

Scenario 1 — Sandbox reset (run before every other scenario)

#	Setup	Action	Expected	Pass?
1.1	—	Debug → Wipe my data	All customers/orders gone from Customers + Dashboard	☐
1.2	—	Debug → Reset Smart usage doc	users/{uid}/usage/smart_drafts deleted in Firestore	☐
1.3	—	Debug → Set tier: Free	Settings → PlanCard shows "Free"	☐
1.4	—	Debug → Reset welcome window (now)	User doc: welcomeBonusAppliedAt ≈ now, bonusCoins = 30	☐

You are now at the "brand-new signed-up tailor" baseline.

Scenario 2 — Welcome window + 30-customer cap

Tests that during the calendar-month welcome window, Free users can add up to **30** customers.

The PlanCard transitions through **three visual states** as you fill the cap:

Customer count	Ratio	State	What you see
0–24 / 30	≤ 80%	Inline (flat card)	FREE pill, X of 30 customers, subtitle "Unlock unlimited customers and reports", small Upgrade > text on the right
25–29 / 30	80%–99%	Warn (amber hero)	Free plan · Almost full pill, N customers left before your free limit title, prominent Upgrade – ₦2,000/mo button + Compare plans →
30 / 30	100%	Locked (red hero)	Free plan · Limit reached pill, You can't add new customers title, prominent Upgrade now button + Compare plans →

#	Setup	Action	Expected	Pass?
2.1	Sandbox reset done	Open Settings → PlanCard	Inline state. FREE pill, counter reads 0 of 30 customers, subtitle "Unlock unlimited customers and reports", Upgrade > text on the right	☐
2.2	—	Debug → Bulk seed → total=24, measurements=10, orders=10 → Seed	All 24 succeed; PlanCard still in inline state (24/30 = 80%)	☐
2.3	—	Add 1 more customer manually (FAB)	Customer count = 25. PlanCard transitions to Warn state: amber hero, Free plan · Almost full pill, title 5 customers left before your free limit	☐
2.4	—	Debug → Bulk seed → total=4, measurements=0, orders=0 → Seed	All 4 succeed; count = 29. Warn-state title now reads 1 customers left... (⚠️ known plural bug — see follow-ups)	☐
2.5	—	Tap FAB → add a 30th customer manually	Save succeeds; count = 30. PlanCard transitions to Locked state: red hero, Free plan · Limit reached pill, title You can't add new customers, Upgrade now button visible	☐
2.6	—	Tap FAB → try to add a 31st	Form blocks with upgrade-prompt copy ("You've reached your customer limit"). Snackbar / dialog references upgrading	☐

2.7	—	Verify Firestore: users/{uid}.customerCount == 30	Server-side counter agrees with UI	☐
2.8	—	Open Settings → PlanCard → tap Upgrade now	UpgradeScreen opens	☐

Why split into 24 → 25 → 29 → 30? The PlanCard states transition at the 80% and 100% thresholds. Splitting the seed lets you verify each transition (inline → warn → locked) in a single test pass. Bulk-seed is additive — second seed adds on top of the first.

Why not just bulk-seed 31? `FirebaseCustomerRepository.createCustomer` enforces the cap on each insert. Bulk-seed silently stops at 30. The manual 31st-tap is what surfaces the upgrade-prompt copy in the form.

Scenario 3 — Welcome window ending warning (3-day banner)

Tests the "Your welcome ends in N days" dashboard banner.

 **Calendar-locked test.** The welcome window expires at midnight on the 1st of the next calendar month (Africa/Lagos). The 3-day warning fires only when today is within 3 days of that boundary. There is **no debug helper** to fake "today" — you can only run this test in the last 3 days of any calendar month, OR by manually backdating `welcomeBonusAppliedAt` in the Firebase Console to a date in last month (which immediately expires the window — see Scenario 4).

#	Setup	Action	Expected	Pass?
3.1	Only runnable in last 3 days of month	Sandbox reset, then open Dashboard	"Your welcome ends in N day(s)" banner appears with Upgrade CTA	☐
3.2	—	Tap CTA	UpgradeScreen opens	☐
3.3	—	Verify <code>welcomeDaysLeft</code> <= 3 in app state	Logs / Reduxomatic shows the value	☐

If you can't run this in the calendar window, document as **skipped** and re-test at month-end.

Scenario 4 — Welcome expired → Free 15-cap + slot reconcile

Tests what happens after the welcome window ends: cap drops to 15, oldest-inactive customers get locked.

#	Setup	Action	Expected	Pass?
4.1	30 customers from Scenario 2, mixed activity	Debug → Bulk seed → total=30, orders=15 (so half have a recent order, half don't)	Seed succeeds; 30 customers total	☐
4.2	—	Debug → Expire welcome window	User doc: <code>welcomeBonusAppliedAt</code> backdated ~35 days	☐

4.3	—	Debug → Reconcile customer slots	Server runs reconcile; snackbar "Slots reconciled"	☐
4.4	—	Open Customers tab	~15 customers grayed out / marked "locked" (the inactive half)	☐
4.5	—	Open Settings → PlanCard	Counter reads 15 / 15 (active), Upgrade CTA visible	☐
4.6	—	Tap a locked customer	SwapSheet opens, lists active customers	☐
4.7	—	Pick an active customer from the sheet	Swap completes; snackbar "Swapped in"; the chosen customer is now locked instead	☐
4.8	—	Try to add a new customer	Upgrade prompt fires (cap = 15)	☐

Note on the 50/50 split. The "active" determination is server-side via `reconcileSlots` and depends on each customer's last-order timestamp. Bulk-seed gives all customers the **same** `createdAt`, so the split depends on which 15 had orders (`orders=15` in 4.1). If your reconcile result looks all-or-nothing, check the cloud function logic — it's server logic and not in scope here.

Scenario 5 — Smart drafts: welcome bonus consumed before free quota

Tests that bonus coins are consumed silently before the visible 5/month free quota.

#	Setup	Action	Expected	Pass?
5.1	Sandbox reset + at least 1 customer with 1 open order	Debug → Set Smart usage → count=0, bonus=30 → Apply	Firestore <code>users/{uid}/usage/smart_drafts</code> has <code>count=0, bonusBalance=30</code>	☐
5.2	—	Smart Suggestions → Draft a Message → pick customer + order + intent → Generate	Draft text appears	☐
5.3	—	Check Firestore	<code>bonusBalance</code> decremented to 29; count still 0	☐
5.4	—	Dashboard chip	Still shows "5 of 5 free drafts left this month" (chip reflects monthly count, not bonus)	☐
5.5	—	Generate 4 more drafts	Bonus drains 29 → 28 → 27 → 26 → 25; count stays 0; chip stays 5/5	☐

Scenario 6 — Smart drafts: free quota exhaustion + upgrade sheet

Tests the transition from bonus → free quota → exhausted.

#	Setup	Action	Expected	Pass?
6.1	Customer + open order ready	Debug → Set Smart usage → count=0, bonus=0 → Apply	Firestore: count=0, bonusBalance=0	☐
6.2	—	Draft a Message → Generate	Draft appears; Firestore count → 1	☐
6.3	—	(Optional, just to refresh the chip) Sign out + sign back in, navigate to Dashboard	Chip reads "4 of 5 free drafts left"	☐
6.4	—	Debug → Set Smart usage → count=4, bonus=0 → Apply	Firestore: count=4	☐
6.5	—	Draft a Message → Generate	Draft appears; count → 5; chip (if refreshed) reads "0 of 5"	☐
6.6	—	Debug → Set Smart usage → count=5, bonus=0 → Apply	Firestore: count=5	☐
6.7	—	Draft a Message → Generate	Modal upgrade sheet opens ("Out of free drafts this month") — NOT inline text	☐
6.8	—	Tap "Upgrade to Pro" on the sheet	UpgradeScreen opens	☐

Inline banner vs modal upgrade sheet. The red inline text above the Generate button (*Out of free drafts this month. Upgrade to keep drafting.*) appears when the **in-memory cache** says `remainingFreeQuota == 0` . The modal bottom sheet only fires when you tap Generate AND the server returns exhausted. They are different surfaces and both can appear in the same session. Trust the modal sheet — it's authoritative.

Scenario 7 — Pro tier (50/month Smart drafts + unlimited customers)

#	Setup	Action	Expected	Pass?
7.1	Sandbox reset	Debug → Set tier: Pro	Settings → PlanCard shows "Pro", customer limit shows "unlimited"	☐
7.2	—	Debug → Bulk seed → total=50, orders=0	All 50 succeed (no cap on Pro)	☐
7.3	—	Debug → Set Smart usage → count=49, bonus=0 → Apply	Firestore: count=49	☐

7.4	—	Generate a draft	Succeeds; count → 50	☐
7.5	—	Debug → Set Smart usage → count=50, bonus=0 → Apply	Firestore: count=50	☐
7.6	—	Generate a draft	Snackbar "You've used your 50 Pro drafts this month" — NOT upgrade sheet (Pro doesn't get a Pro→Pro upgrade prompt)	☐

Pro hits **pro_quota_exhausted** server-side, which the client routes to a snackbar instead of the upgrade sheet. This is the tier-aware copy that ships with V1.0.

Scenario 8 — Atelier tier (unlimited everything)

#	Setup	Action	Expected	Pass?
8.1	Sandbox reset	Debug → Set tier: Atelier	Settings → PlanCard shows "Atelier"; unlimited customers	☐
8.2	—	Debug → Bulk seed → total=100, orders=0	All 100 succeed (Pro/Atelier have no cap)	☐
8.3	—	Generate Smart drafts repeatedly (e.g. 10 of them, or set count=1000)	All succeed; no quota exhaustion at any number	☐
8.4	—	Check Firestore	usage/smart_drafts may or may not exist (Atelier bypasses the reservation transaction)	☐

Scenario 9 — Upgrade flow

#	Setup	Action	Expected	Pass?
9.1	Free + cap-hit state (Scenario 2 step 2.6)	Tap upgrade CTA in PlanCard	UpgradeScreen opens with Pro + Atelier options	☐
9.2	—	Pick Pro → Continue	Paystack web view opens (test mode)	☐
9.3	—	Complete Paystack test transaction	Returns to app	☐
9.4	—	Wait for entitlements to refresh OR	Tier upgrades to Pro automatically (V1.0 manual server flip if Paystack webhook not wired)	☐
9.5	—	Verify Settings hero badge	Shows "Pro" badge on profile hero	☐

Scenario 10 — Account switch regression

Quick check that signing out + back in doesn't leak state.

#	Setup	Action	Expected	Pass?
10.1	Active session as Fola with non-zero data	Debug → Switch to Gabby	Login + onboarding skip; Gabby's customers shown	☐
10.2	—	Open Smart → Draft a Message	Dashboard chip is hidden (Gabby's in-memory cache is null)	☐
10.3	—	Debug → Switch to Fola	Back to Fola; her customer count + chip restore	☐

Known V1.0 quirks (NOT bugs)

These are expected behaviors that may look confusing during testing:

- **Smart chip lag.** The dashboard "X of 5 free drafts" chip reads from a **process-local in-memory cache** that only updates after a real successful draft. If you use `Set Smart usage` debug action to change the Firestore doc, the chip won't reflect it until you (a) sign out + back in, or (b) generate one successful draft. **The server is still authoritative** — the upgrade sheet fires correctly even when the chip is stale. V1.1 will add Firestore-backed chip hydration.
- **Welcome 3-day warning calendar-locked.** As noted in Scenario 3, this only fires when today is within 3 days of the next-month boundary. No debug shortcut without also mocking the clock (out of scope for V1.0).
- **Drain bonus coins** writes to the user-doc `bonusCoins` field (the seed), NOT to the `bonusBalance` on the usage doc (the lifted balance). Use **Set Smart usage** to control the lifted balance the server actually checks during Smart calls.
- **Bulk-seed is additive.** It doesn't wipe — pair with `Wipe my data` first if you need a clean slate. Subject to current tier cap (won't seed past 30 on Free welcome).
- **Welcome 30-cap collisions.** If you've already added customers and then `Reset welcome window`, the cap remains at 30. To re-test the cap-hit copy, use `Wipe my data` first.

When tests fail — diagnostics

Symptom	First place to look
Something went wrong. Please try again on draft	Firebase Functions logs (<code>smartDraftMessage</code>) — usually a Vertex AI error or a Firestore field validation error
Customer cap not enforced	Check <code>users/{uid}.subscriptionTier</code> in Firestore — must be lowercase "free", not "FREE"
Locked customers don't appear after expire	Confirm <code>reconcileSlots</code> ran (debug action snackbar) — check Functions logs

Upgrade sheet doesn't fire	Check Functions logs for permission-denied + the marker (<code>free_tier_exhausted</code> or <code>pro_quota_exhausted</code>) — must match client's <code>SmartFunctionsRepository.recoverFromMessage</code> markers
Smart chip wrong number	Confirmed quirk above. Source of truth is Firestore <code>usage/smart_drafts.count</code>

Going from this doc to a Google Sheet

To convert to a Google Sheet:

1. Copy each table block.
2. Paste into a Google Sheet — the columns will auto-split.
3. Add a "Tester", "Date", "Notes" column on the right.
4. Use checkbox cells for the Pass column (`Insert` → `Checkbox`).

To track across multiple test runs, duplicate the sheet per release candidate and date it.